Daily Drilling Report
SPD19B-03
9/7/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
42.0	8 1/2"	187.00	72.83	31.70	91.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	213,504.94	37,337.50	10,297,226.93		37.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,842.00	2,940.00		98.00	21.77	4.5

Operations at Report Time
Drilling @ 2949m.

Operations Summary
Directional drilling 8 1/2" hole section f/ 2842m to 2940m.

Operations next 24hrs
Cont' drilling 8 1/2" hole section. Stop drilling due to low ROP. Pump out of the hole. RIH New BHA#11.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Behnam Jahangiri	Night Drilling Sup.
Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	9/4/2014	9/26/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	9/6/2014	10/7/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	9/6/2014	10/22/2014
Trip Drill	9/4/2014	10/25/2014

STOP Cards

Number of Reports: 33

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00	24....	8 1/2"	D	03	P	Directional drilling 8 1/2" hole section in rotation mode f/ 2842m to 2940m w/ 570 gpm /2300-2350 psi / 25-35 kips/ 60 rpm / 14-22 Kft-lbf. ** Drilling with NL/NG. ** B/R & R/D each stand drilled once. ** Pump 20 bbl Hi-Vis pill when std down. ** Take survey before connection.
00:00	00:00		8 1/2"				Sep 08, 2014 00:00-06:00 Cont' directional drilling 8 1/2" hole section in rotation mode f/ 2940m to 2949m w/ 570 gpm / 2300-2350 psi / 35 kips / 60 rpm / 14-22 Kft-lbf. ** Drilling with NL/NG. ** B/R & R/D each stand drilled once. ** Take survey before connection. >> Last sample @ 2945m: Lst (30 %) + Dol. (70 %)

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,940.00	CaCo3/KCL/ Polymer	10.50	55	27.0	21.000	2.7
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.0	0.500	1.500	1.300			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	7	0.0	87.0	13.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
6.000	8.000	10.000	0/100	61.0	1424.0	2800.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
7349.0	124.0	0.0	2.26		12.55214	0.435

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.20	6.	300	4.0	998.0	36.7

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Mataf Pride	Crew Boat	9/7/2014 21:40	9/7/2014 21:55	Arrived from Sahar 2, Embarked 01 PAX for Kish, departed to Kish via Sahar-1.
Ahmed 6	Supply Boat	9/7/2014 17:30		Arrived from Sahar 2, Delivered Cargo, Proceed to Anchorage & Standby.

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Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Mouje Darya	Supply Boat	9/8/2014 01:30		Arrived from Kish , Secured A/S at 0200 Hrs. Delivering cargo (23 PPF 7" TBG) in progress.

Mud Additives

Description	Consumed	Cum Rec	On Loc
CaCL2 POWDER 94-97%	50.0	50.0	0.0
Duo-Tec	7.0	185.0	113.0
Lime 2014	30.0	120.0	90.0
Starch	20.0	1,600.0	377.0
Soda Ash 25 Kg	30.0	750.0	311.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Drill Water	3.0		320.0
Potable Water	21.0	18.0	199.0
Cement G	0.0	0.0	92.0
Diesel Fuel	15.0		173.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	129.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
10	8 1/2	Varel	VM713GUX	6006867	PDC	M332	14/14/14/14
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.60	52.67	686,699.40	2,725.00	2,940.00	215.00	4.1	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
500	575	1,825.0	2,400.0	15	35	160	242
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
42.47	49.27	93.54	6.46	443	365	285	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,842.00	2,872.00	8.07	19,000.0	570	60	2,380.0	25	360
2,872.00	2,940.00	13.70	19,000.0	570	60	2,400.0	35	365

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.89	255.95
7" Drilling Jar (COR-3021AD)	7	2.938	5.80	228.06
20 jts 5 7/8" HWDP	5 7/8	4.000	186.35	222.26
X/O	6 1/2	2.813	0.85	35.91
6 3/4" Float sub	6 3/4		0.81	35.06
6 3/4" NMDC	6 3/4	3.250	9.48	34.25
6 3/4" TM - Sub	6 3/4		1.74	24.77
6 3/4" NMDC	6 3/4	3.250	9.48	23.03
8" St. Stabilizer (COR-2491AD)	8	3.000	1.84	13.55
Pony NMDC	6 13/16	3.500	3.09	11.71
6 3/4" Motor (AKO: 1.15° , Sleeve: 8 3/8")	6 3/4		8.34	8.62
8 1/2" PDC bit	8 1/2		0.28	0.28

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
1	2,865.00	14.00	20	220.0	97	114
1	2,865.00	14.00	30	345.0	97	170
1	2,865.00	14.00	40	550.0	97	227
2	2,865.00	14.00	20	215.0	97	114
2	2,865.00	14.00	30	340.0	97	170
2	2,865.00	14.00	40	540.0	97	227

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
9/5/2014	9 5/8" Shoe, 2,720.80mKB	Formation Integrity	2,728.00	2,583.00	1,540.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,540.0	5.7	1,540.00		14.00	6,162.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,845.41	251.40	25.60	2,690.79	-356.67	-792.01	868.62	0.14	0.00
2,874.50	251.60	25.60	2,717.02	-360.65	-803.93	881.13	0.09	0.00
2,903.03	251.40	26.20	2,742.69	-364.61	-815.75	893.53	0.64	0.63

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Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S1		2,675.70	2,534.34
Upper Kangan		2,703.00	2,559.14
K1		2,732.00	2,586.77
K2		2,851.80	2,696.47
K3		2,901.00	2,740.86

General Remarks

Comment

>> Requirements: (200 sxs Starch f/ MR #14)

>>> Sent:

** Nil

>>> Received:

** Ahmed 6 f/ SR2: (Alghaith side door elevator) + (ABAN Gas Cylinder Rack)